

BAHRAIN INTL, Bahrain

WMO: 411500

Lat: 26.2618N

Lon: 50.6433E

Elev: 6

StdP: 14.69

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	54.4	56.8	35.3	30.2	67.9	38.9	34.9	69.4	28.1	60.3	26.3	61.0	13.7	310	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	106.1	74.6	104.2	76.1	102.3	77.2	88.1	95.9	87.2	95.3	86.2	94.8	10.1	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
86.2	191.9	94.3	85.1	184.9	93.9	84.1	178.9	93.6	53.4	96.3	52.3	94.8	51.0	94.5	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.9	20.8	DB	50.5	112.3	3.0	1.8	48.4	113.6	46.6	114.7	44.9	115.7	42.7	117.0	(4)
(5)			WB	44.1	89.4	3.6	1.6	41.5	90.5	39.4	91.5	37.4	92.4	34.8	93.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.8	64.1	65.9	71.5	79.8	89.3	94.0	96.2	96.2	92.6	86.6	76.7	67.7	(6)
(7)		DBStd	12.38	4.19	4.27	4.67	4.72	3.98	3.13	2.55	2.61	3.03	3.33	4.85	4.46	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	131	68	35	4	0	0	0	0	0	0	0	0	24	(9)
(10)		CDD50	11602	436	445	667	894	1218	1319	1432	1432	1278	1133	800	548	(10)
(11)		CDD65	6258	39	60	206	444	753	869	967	967	828	668	350	107	(11)
(12)		CDH74	86699	30	85	769	4031	10869	14050	16087	15977	13020	9015	2594	172	(12)
(13)		CDH80	54930	1	9	180	1392	6489	9731	11623	11513	8704	4674	606	8	(13)
(14)	Wind	WSAvg	10.1	10.9	11.5	10.9	10.0	10.1	11.4	10.3	8.6	8.6	8.3	10.1	10.5	(14)
(15)	Precipitation	PrecAvg	2.3	0.6	0.6	0.4	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.4	(15)
(16)		PrecMax	6.5	3.5	4.2	2.8	2.0	0.4	0.0	0.2	0.0	0.0	1.2	2.4	3.8	(16)
(17)		PrecMin	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.6	0.7	0.9	0.6	0.4	0.1	0.0	0.0	0.0	0.0	0.2	0.5	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.2	80.7	91.0	98.2	105.9	109.1	109.8	109.1	105.4	99.0	91.1	80.7	(19)
(20)			MCWB	62.3	62.6	64.5	68.0	69.7	70.6	73.4	75.9	74.6	74.3	69.6	67.4	(20)
(21)		2%	DB	73.7	76.7	84.5	93.2	102.1	105.4	106.0	105.7	102.1	95.9	87.5	77.4	(21)
(22)			MCWB	62.9	62.5	64.3	68.4	70.5	72.0	75.2	78.0	77.5	75.6	72.2	67.5	(22)
(23)		5%	DB	71.7	73.5	80.6	89.7	98.8	102.4	103.9	103.6	100.2	93.6	85.6	75.5	(23)
(24)			MCWB	63.4	62.8	64.6	68.2	71.5	73.7	76.5	79.8	78.7	76.2	72.0	67.1	(24)
(25)		10%	DB	69.9	71.5	77.4	87.0	96.5	100.3	101.9	101.8	98.5	91.8	83.6	73.8	(25)
(26)			MCWB	62.7	62.8	64.7	68.7	71.9	74.9	77.9	80.9	79.3	76.5	71.8	66.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	68.9	71.8	75.8	82.5	85.5	89.1	89.8	87.9	83.8	78.4	72.9	(27)
(28)			MCDWB	71.9	73.1	78.8	85.6	90.1	93.6	95.7	97.4	96.0	91.1	84.3	76.7	(28)
(29)		2%	WB	67.0	67.2	69.9	74.4	80.2	83.6	87.7	88.5	86.4	82.3	76.4	70.6	(29)
(30)			MCDWB	70.8	71.6	76.0	84.2	89.4	92.4	95.2	96.4	94.9	89.8	83.4	74.8	(30)
(31)		5%	WB	65.3	65.9	68.8	73.1	78.8	82.1	86.6	87.7	85.2	81.0	74.9	69.1	(31)
(32)			MCDWB	69.9	70.5	75.2	83.1	89.0	91.8	94.5	95.8	94.0	88.8	82.2	73.8	(32)
(33)		10%	WB	63.7	64.8	67.3	72.0	77.2	80.7	85.4	86.8	84.2	80.0	73.6	67.5	(33)
(34)			MCDWB	68.5	69.8	74.4	82.1	88.9	92.4	94.1	95.3	93.2	88.0	80.8	72.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	11.3	12.6	13.8	13.0	12.4	12.0	12.1	11.3	9.3	9.0	(35)
(36)			MCDBR	11.1	12.8	16.2	16.9	17.7	16.3	15.7	14.7	13.7	13.1	11.0	10.2	(36)
(37)			MCWBR	6.3	6.4	7.5	8.3	9.6	10.4	11.1	10.1	9.4	8.4	7.5	6.2	(37)
(38)		5% WB	MCDWB	9.1	10.3	12.4	14.4	15.2	14.6	13.0	12.2	11.9	12.0	10.2	9.0	(38)
(39)			MCWBR	6.7	6.7	7.5	8.1	9.7	10.2	9.9	8.6	8.0	7.8	7.6	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.509	0.557	0.619	0.678	0.669	0.630	0.676	0.633	0.559	0.520	0.522	0.492	(40)	
(41)		taud	1.937	1.827	1.694	1.625	1.630	1.698	1.657	1.752	1.879	1.963	1.959	1.995	(41)	
(42)		Ebn at Noon	228	228	223	214	216	223	213	222	235	236	223	227	(42)	
(43)		Edh at Noon	53	63	76	83	83	77	80	72	62	54	51	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1146	1378	1656	1851	2131	2298	2163	2053	1902	1584	1190	1083	(44)	
(45)		RadStd	78	91	120	124	64	61	49	43	38	47	82	81	(45)	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	+1.06	N/A	-2.59	+1.31	N/A	N/A	N/A	+425	+403	(46)
(47)	Regional (5 neighbors)	+0.92	N/A	-2.74	+1.26	N/A	N/A	N/A	+353	+344	(47)

Nomenclature: See separate page